

ANUSHKA NAYAK

anushkanayak15@g.ucla.edu | (+1) 424-857-3599 | Los Angeles, CA | www.linkedin.com/in/anushkanayak15 | <https://github.com/anushkanayak15>
Software engineer building AI-powered products, with applied ML research in LLM fine-tuning, RAG, and model evaluation, experimentation and growth engineering projects, and production experience at American Express and EY.

EDUCATION

University of California, Los Angeles (UCLA), Henry Samueli School of Engineering
Bachelor of Science in Computer Science

Jun 2026

- **Relevant Coursework:** Algorithms and Data Structures, OS Principles, Computer Architecture, Database Systems, Software Engineering, Distributed Systems, Machine Learning, Artificial Intelligence, Data Mining, Probability, Discrete Math.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Frameworks and Databases: Next.js, React, Node.js, Flask, REST APIs, PostgreSQL, MySQL, Firebase

AI / Machine Learning: PyTorch, Hugging Face Transformers, LoRA fine-tuning, RAG, FAISS, embeddings, vector search, LLM APIs, prompt engineering, model evaluation, scikit-learn, TensorFlow

Cloud & DevOps: AWS, Google Cloud (GCP), Docker, CI/CD, Linux/Unix, Git/GitHub, Gerrit

EXPERIENCE

American Express • Software Engineering Intern, Distributed Infra and Data Svc Team

Jun 2025 – Aug 2025

- Designed and deployed a production Flask + PostgreSQL API replacing a legacy DNS platform, **reducing provisioning time by 70%** and **supporting 100+ automated requests/day**.
- Defined API requirements and documented **11 RESTful endpoints** in Swagger, improving scalability and cross-team adoption.
- Shipped 3 production features automating DNS record lifecycle workflows, eliminating manual provisioning tickets for teams.

Ernst & Young • Data Analysis and Solutions Intern

Jul 2024 - Sep 2024

- Developed **5 Python automation** pipelines for reporting and processing large client datasets, saving **15+ hours/week** and improving **team productivity by 35%**.
- Trained regression, clustering, and decision tree models on 200K+ records to inform client resource allocation strategy.
- Hardened pipelines with preprocessing and validation layers, reducing downstream reporting errors.

Boerick Research Laboratory • Business Intelligence Analyst Intern

Jun 2023 - Sep 2023

- Built Tableau dashboards integrated with SQL data sources to track supply chain KPIs, reducing **overstocking costs by 15%**.
- Revamped e-commerce platform using HTML, CSS, and JS, boosting **engagement by 20%** and reducing **bounce rate by 18%**.

PROJECTS

GrowthOS: AI-Powered Landing Page Optimization Engine • Next.js, TypeScript, LLM APIs

Jun 2026 - Jul 2026

- Architected and deployed a full-stack experimentation platform using Next.js and TypeScript that generates, evaluates, and ranks AI-created landing page variants through an automated optimization pipeline.
- Built a behavioral simulation and analytics engine to model user personas, evaluate landing page variants, and identify top-performing designs through offline experimentation.

Fairness-Aware ASR Fine-Tuning (Research, UCLA) • Hugging Face, LoRA, Whisper, FLEURS

May 2026 - Jun 2026

- Fine-tuned **Whisper using LoRA** on FLEURS dataset and engineered an end-to-end pipeline for fairness-aware ASR evaluation.
- Designed a framework to measure **gender-based WER** disparities and compare baseline versus fine-tuned model performance.

sEMG-Based Keystroke Decoding System • Team Lead • PyTorch, PyTorch Lightning, CNN, BiLSTM, CTC

Feb 2026 - Mar 2026

- Implemented CNN + BiLSTM + CTC model for sEMG-to-text decoding, **reducing CER from 30% to 15.5%** vs baseline.
- Built spectrogram-based preprocessing and augmentation pipeline to improve model generalization and reduce validation CER.
- Developed GPU-accelerated PyTorch Lightning training pipeline and tuned hyperparameters to outperform baseline models.

Gender Bias in Language Models • Team Lead • Python, FastText, Gensim, NLP, WEAT

Jan 2026 - Feb 2026

- Analyzed gender bias across job seniority levels using FastText embeddings and WEAT effect size metrics.
- Built end-to-end NLP pipelines for corpus cleaning, vocabulary filtering, and permutation-based significance testing.

LEADERSHIP and INVOLVEMENT

UCLA Computer Science Department • Undergraduate Reader

Jul 2025 - Jun 2026

- Defined grading standards and partnered with faculty to streamline evaluation of programming labs for **100+ students**, **improving instructional efficiency**, consistency of feedback and overall learning outcomes.

Association for Computing Machinery (ACM-W UCLA) • External Officer

Sep 2024 - Jun 2026

- Led and moderated industry panel with FAANG speakers, engaging **60+ attendees** through curated speaker lineup.

National Students Data Corps UCLA • Board Advisor

Sep 2023 - Jun 2026

- Scaled membership from **50 to 850+** through data-driven growth initiatives and strategic outreach campaigns.
- Led website redesign resulting in **30% traffic increase within 2 months** through improved UX and social-media strategy.